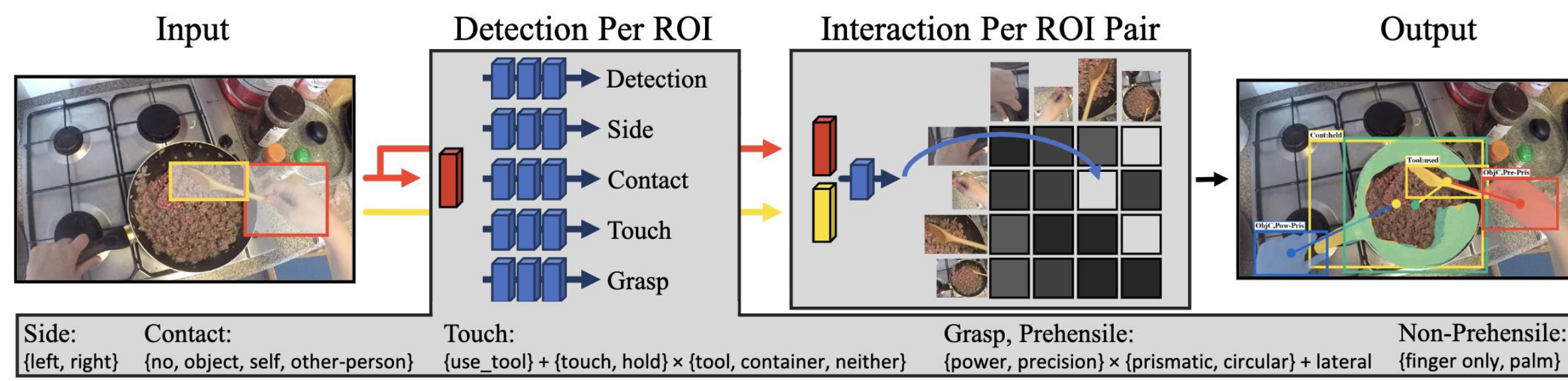
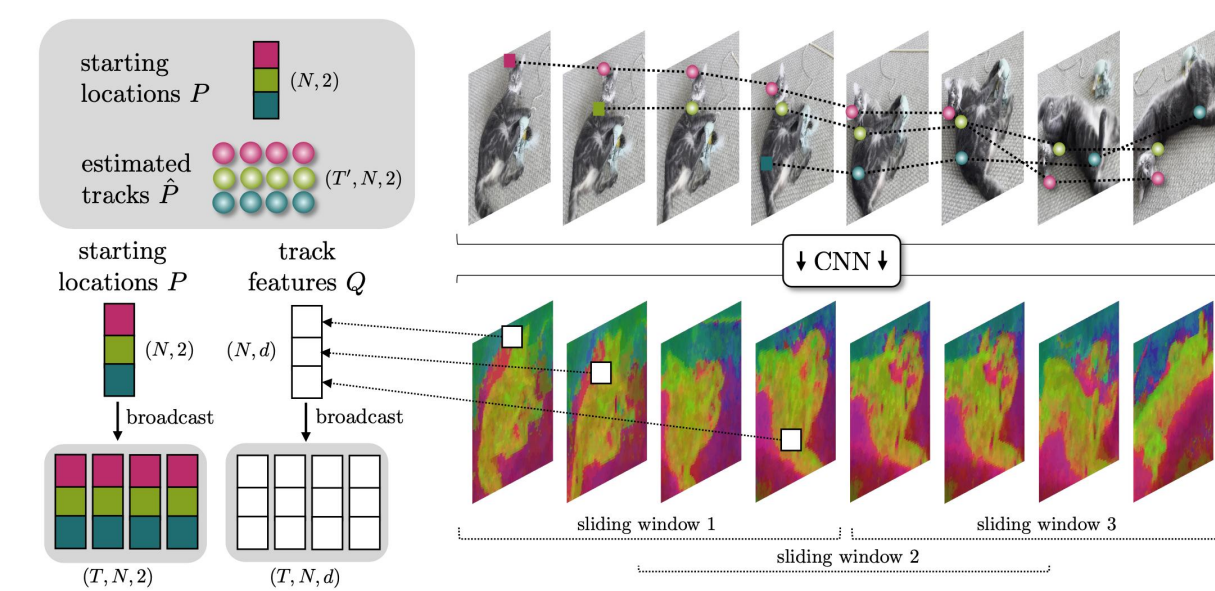


2D Space

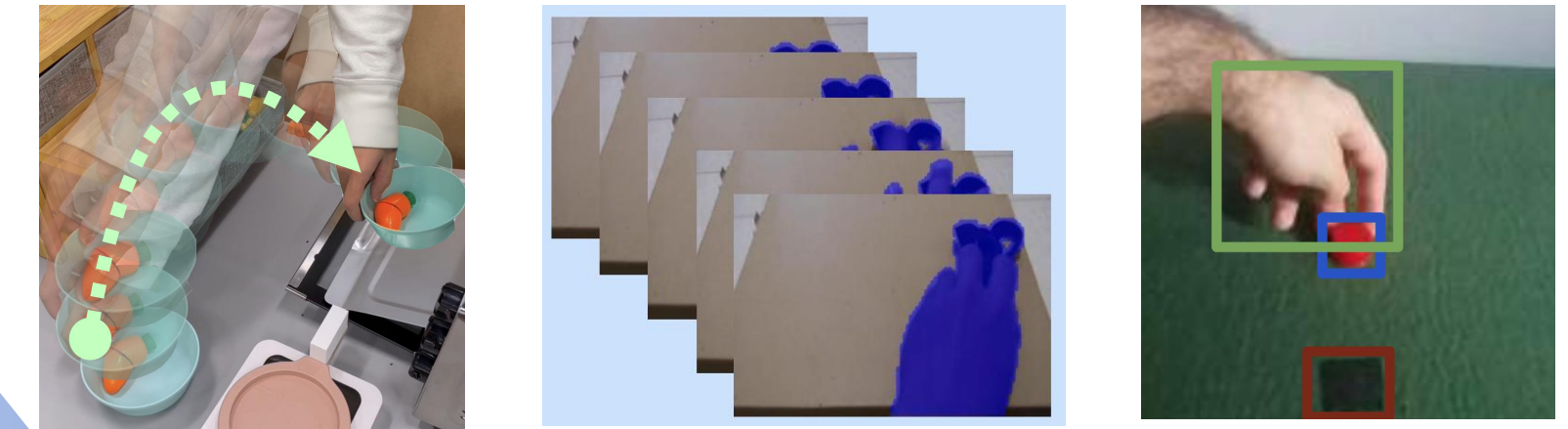


(a) 2D HOI Detection

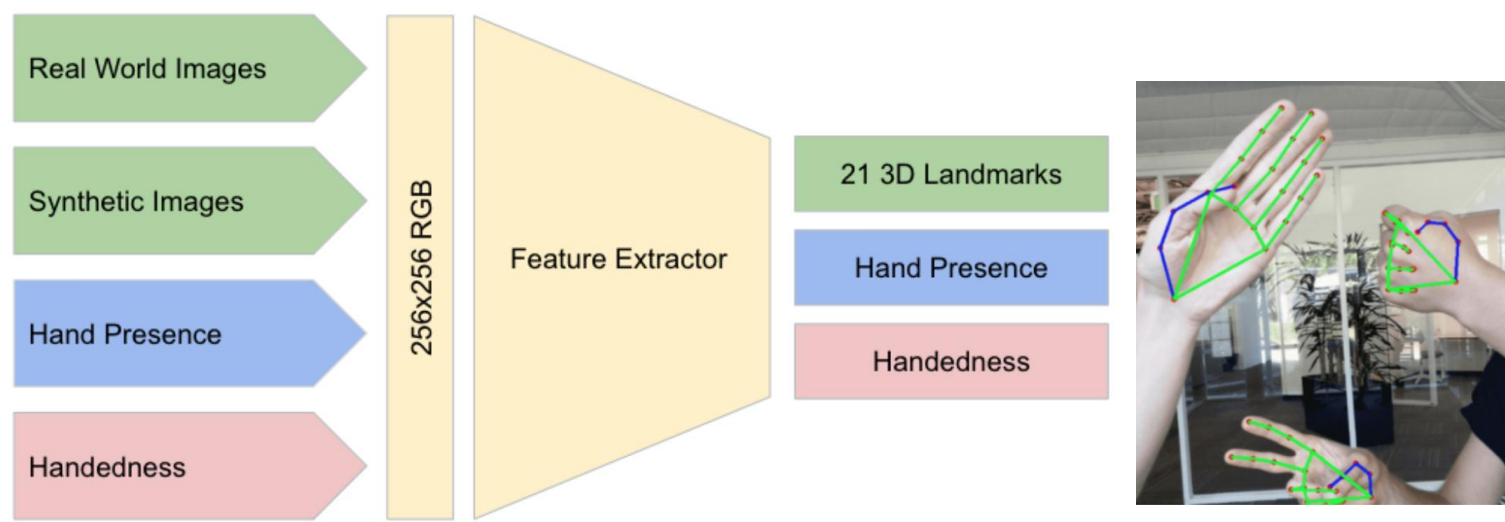


(b) 2D Point Tracking

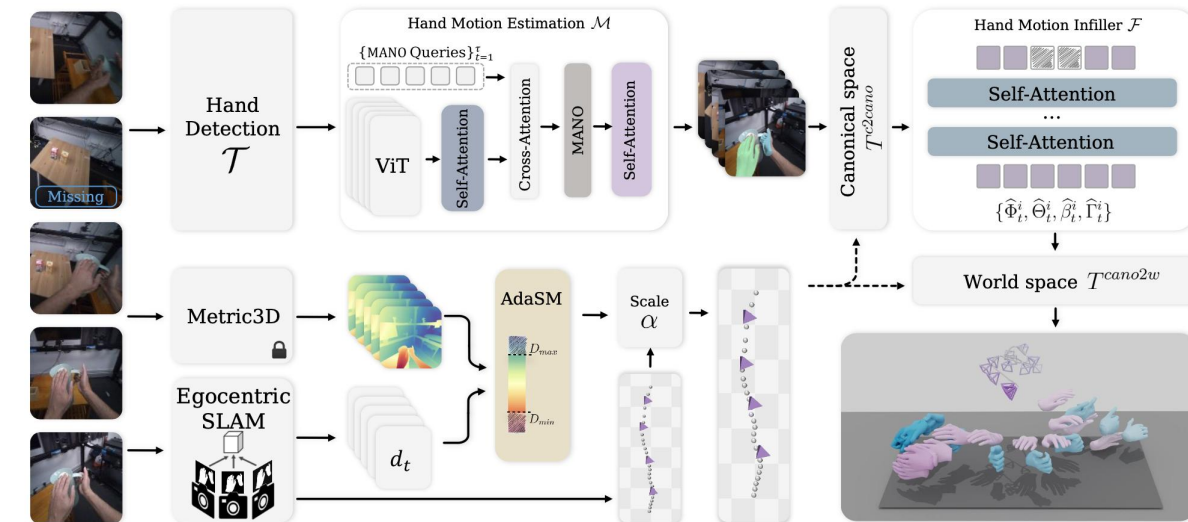
Affordances



2D/3D Hand&Object Positions

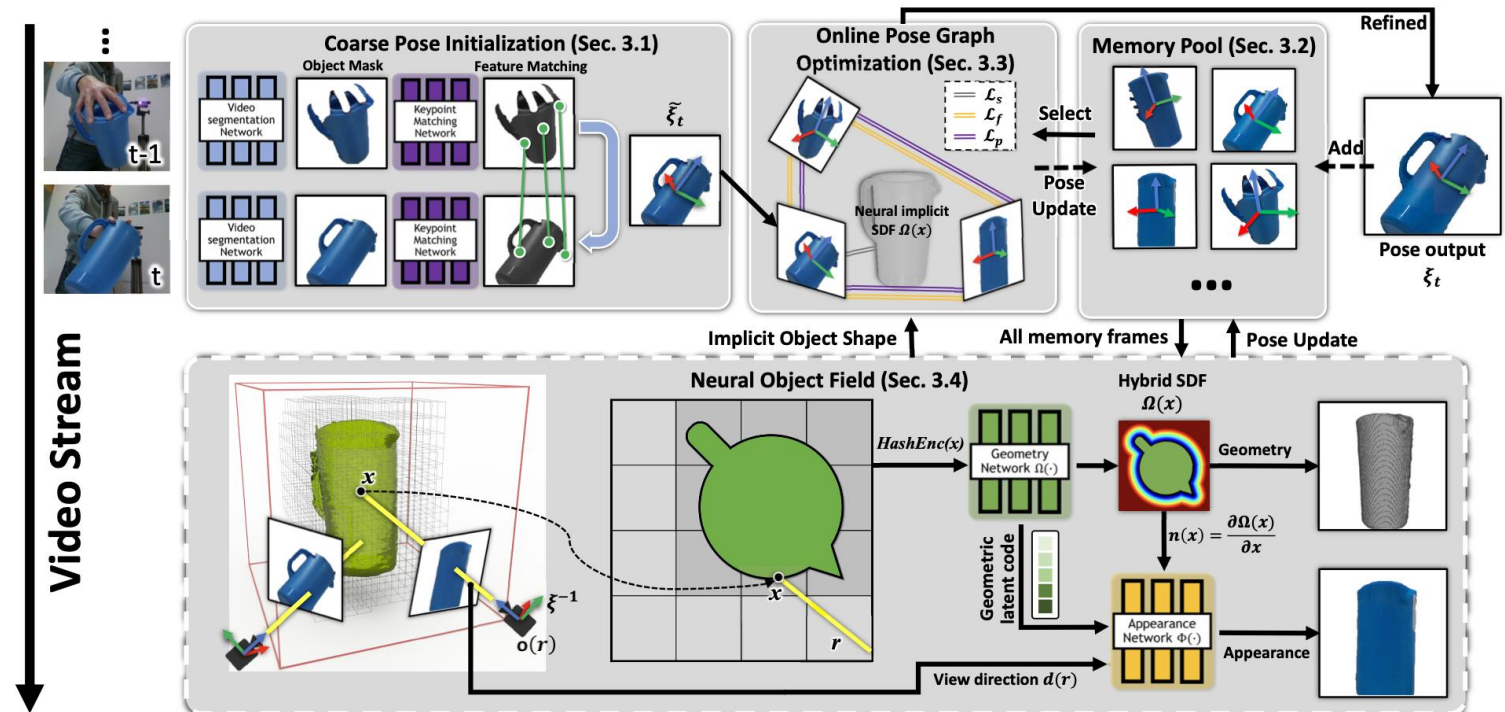


(c) 3D Hand Landmark Detection

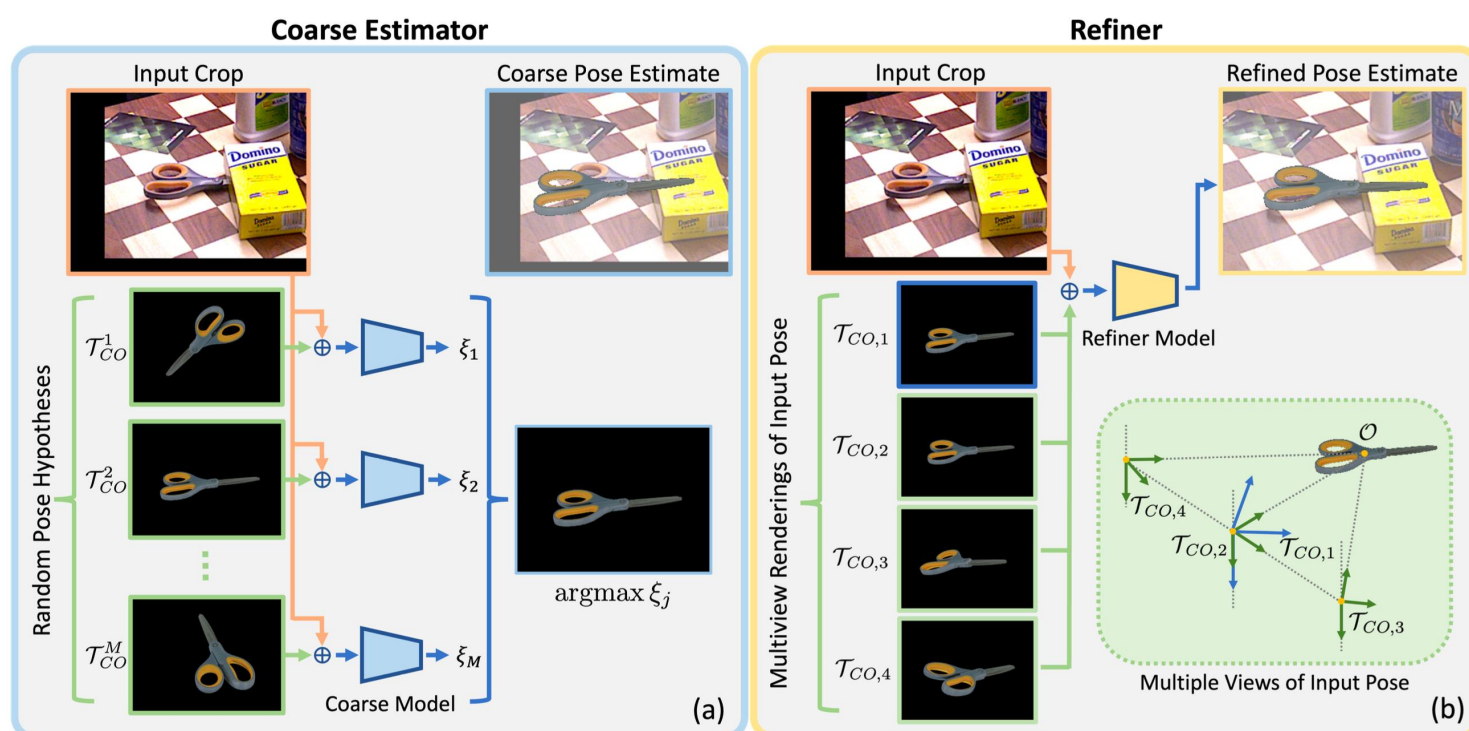


(d) 3D Hand Reconstruction

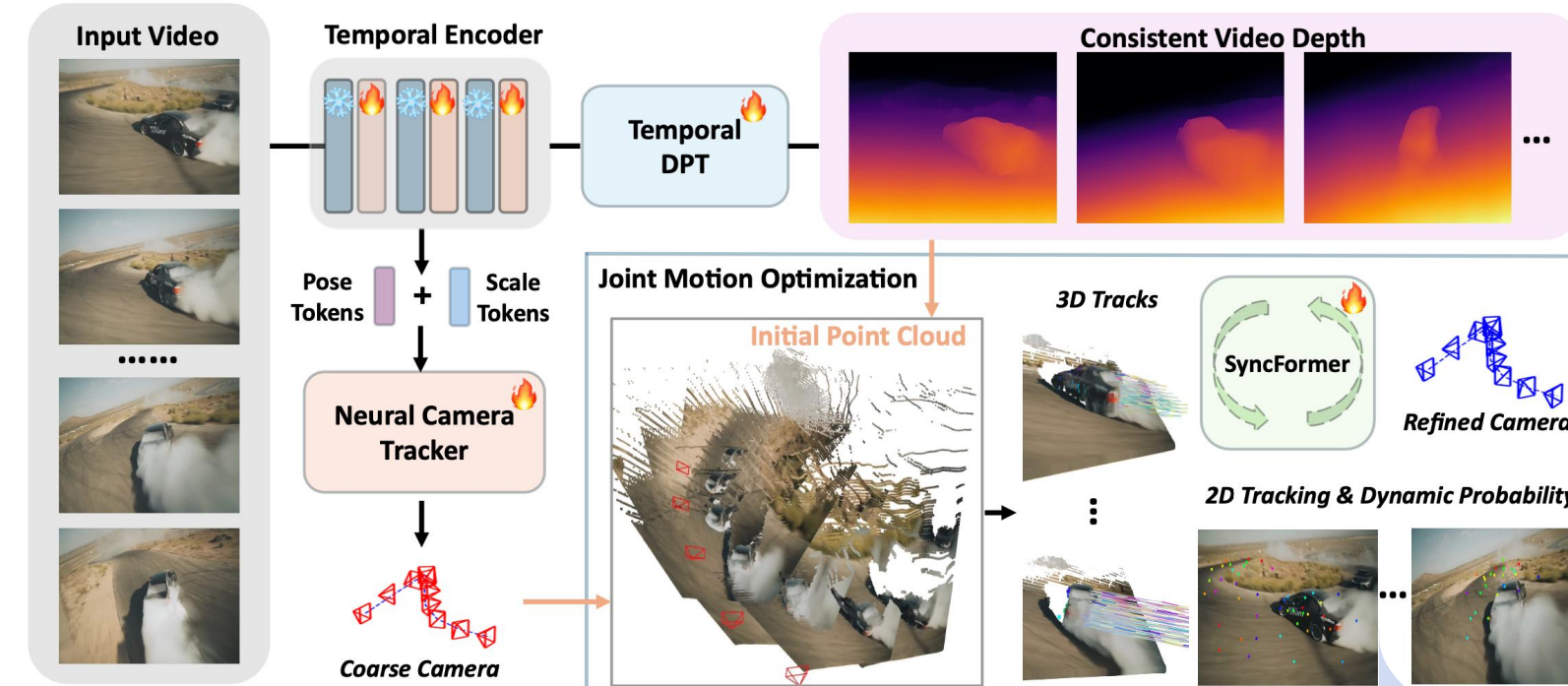
3D Space



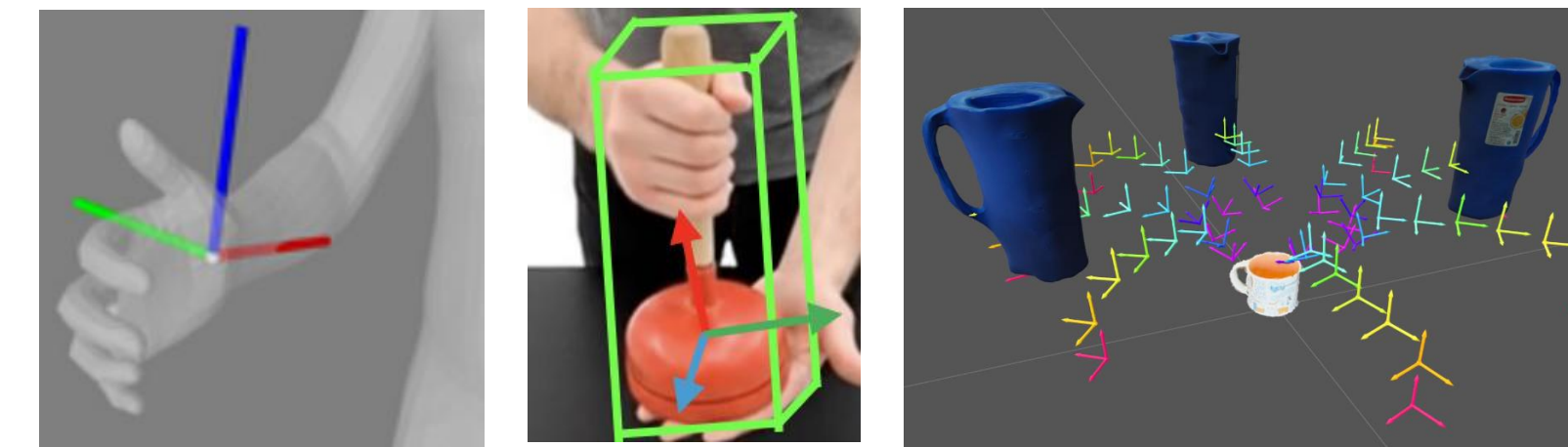
(e) 3D Object Reconstruction



(f) 6D Object Pose Estimation

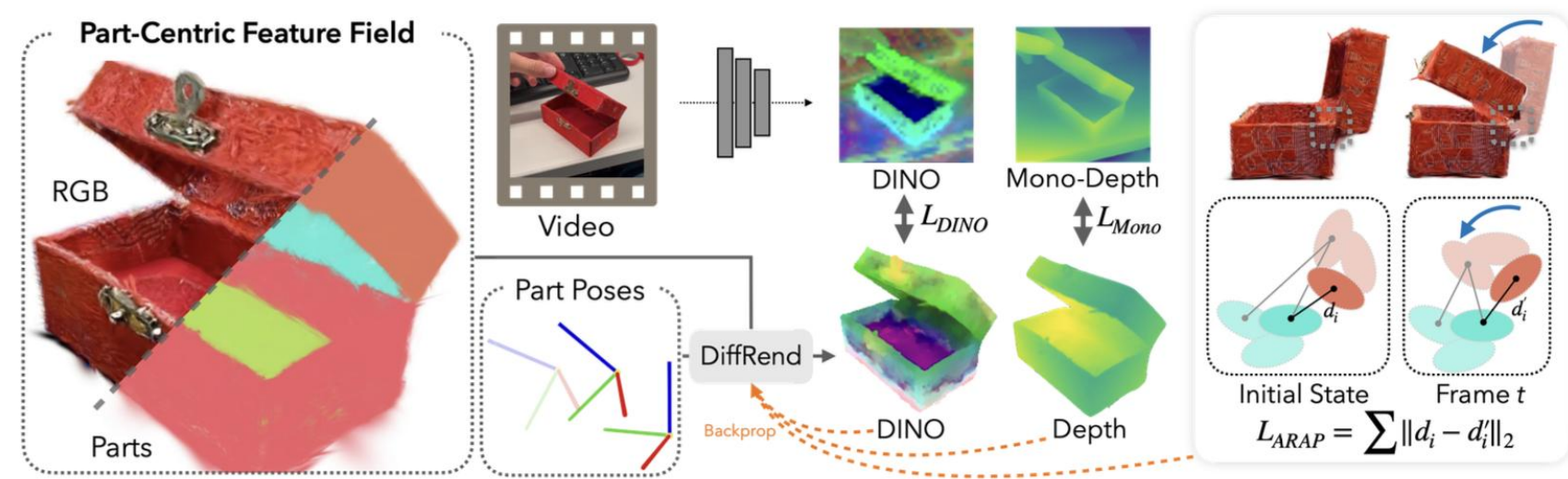


(g) 3D Point Tracking

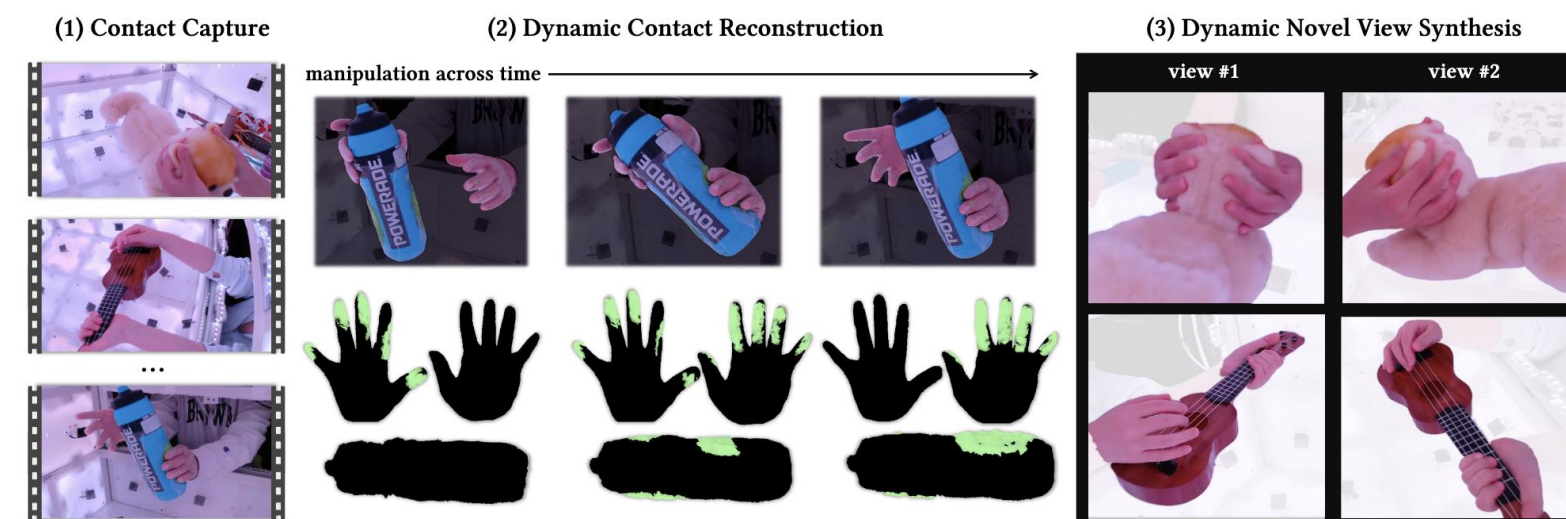


6D Hand&Object Poses

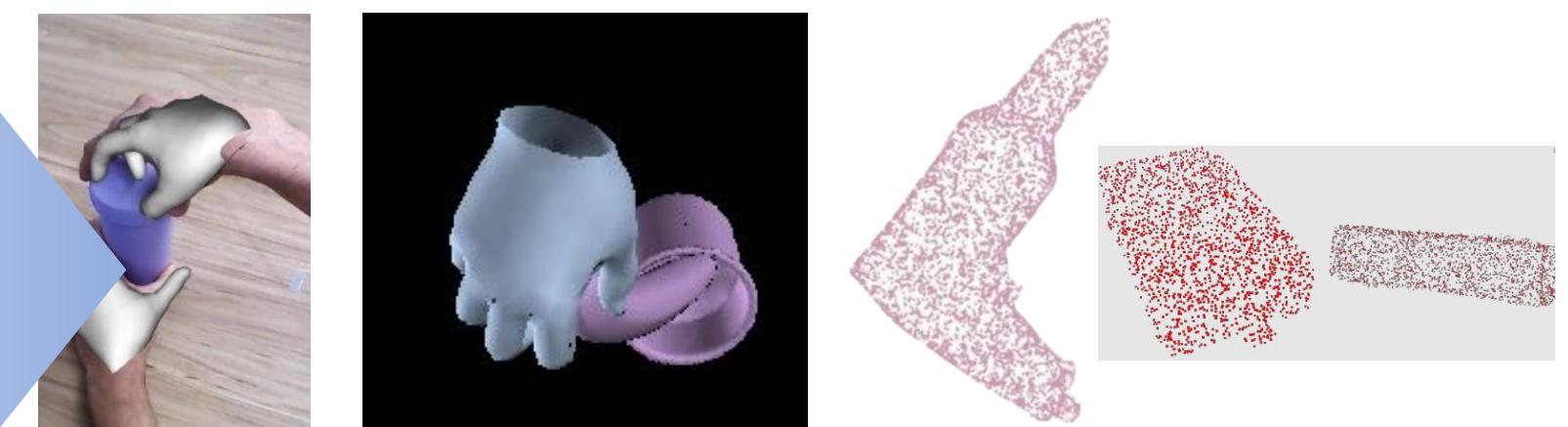
4D Space



(h) 4D Object Reconstruction



(i) 4D Contact Evolution Reconstruction



Hand&Object Geometric Representations



Physical Object Functionality